LAB ASSIGNMENT – 1
DATE – 05.01.2026

NAME : Soma Chatterjee
REGISTRATION NO. : AU/2023/0009439

ROLL NO. : UG/02/BTCSECSF/2023/013

SUBJECT : Unstructured Database Lab
SEC: E

Q1. Create a database named collegeDB and switch to it.

```
mongosh mongodb://127.0.0.
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Soma Chatterjee>mongosh
Current Mongosh Log ID: 69d2a598363c60005a7c2906
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:          8.2.5
Using Mongosh:          2.7.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-04-04T20:26:52.211+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> use collegeDB
switched to db collegeDB
```

Q2. Create a collection named students in the collegeDB database.

```
collegeDB> db.createCollection("students")
{ ok: 1 }
```

Q3. Insert one document into the students collection with the following fields:

- Name
- rollNo

- department
- age

```
collegeDB> db.students.insertOne({
|    name: "Soma Chatterjee",
|    rollNo: 101,
|    department: "CSE",
|    age: 20
| })
{
  acknowledged: true,
  insertedId: ObjectId('69d2a63b363c60005a7c2907')
}
```

Q4. Insert four more documents into the students collection with different values.

```
collegeDB> db.students.insertMany([
|    { name: "Amit Sharma", rollNo: 102, department: "ECE", age: 21 },
|    { name: "Priya Das", rollNo: 103, department: "CSE", age: 19 },
|    { name: "Rahul Sen", rollNo: 104, department: "ME", age: 22 },
|    { name: "Neha Roy", rollNo: 105, department: "CSE", age: 20 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69d2a64d363c60005a7c2908'),
    '1': ObjectId('69d2a64d363c60005a7c2909'),
    '2': ObjectId('69d2a64d363c60005a7c290a'),
    '3': ObjectId('69d2a64d363c60005a7c290b')
  }
}
```

Q5. Display all documents present in the students collection.

```
collegeDB> db.students.find()
[
  {
    _id: ObjectId('69d2a63b363c60005a7c2907'),
    name: 'Soma Chatterjee',
    rollNo: 101,
    department: 'CSE',
    age: 20
  },
  {
    _id: ObjectId('69d2a64d363c60005a7c2908'),
    name: 'Amit Sharma',
    rollNo: 102,
    department: 'ECE',
    age: 21
  },
  {
    _id: ObjectId('69d2a64d363c60005a7c2909'),
    name: 'Priya Das',
    rollNo: 103,
    department: 'CSE',
    age: 19
  },
  {
    _id: ObjectId('69d2a64d363c60005a7c290a'),
    name: 'Rahul Sen',
    rollNo: 104,
    department: 'ME',
    age: 22
  },
  {
    _id: ObjectId('69d2a64d363c60005a7c290b'),
    name: 'Neha Roy',
    rollNo: 105,
    department: 'CSE',
    age: 20
  }
]
```

Q6. Display only those students who belong to the CSE department.

```
collegeDB> db.students.find({ department: "CSE" })
[
  {
    _id: ObjectId('69d2a63b363c60005a7c2907'),
    name: 'Soma Chatterjee',
    rollNo: 101,
    department: 'CSE',
    age: 20
  },
  {
    _id: ObjectId('69d2a64d363c60005a7c2909'),
    name: 'Priya Das',
    rollNo: 103,
    department: 'CSE',
    age: 19
  },
  {
    _id: ObjectId('69d2a64d363c60005a7c290b'),
    name: 'Neha Roy',
    rollNo: 105,
    department: 'CSE',
    age: 20
  }
]
```

Q7. Update the age of one student (any one) to a new value.

```
collegeDB> db.students.updateOne(
|    { rollNo: 101 },
|    { $set: { age: 21 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Q8. Delete one document from the students collection using any condition

```
collegeDB> db.students.deleteOne({ rollNo: 104 })
{ acknowledged: true, deletedCount: 1 }
collegeDB>
```

MongoDB Compass - localhost:27017/collegeDB.students

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

- localhost:27017
  - admin
  - collegeDB
    - students
  - config
  - libraryDB_RollNo
  - local
  - skillforge
  - test
- test

localhost:27017 | students | My Queries

localhost:27017 > collegeDB > students

Open MongoDB shell

Documents 0 | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or Generate query

Explain | Reset | Find | Options

ADD DATA | UPDATE | DELETE | EXPORT DATA | EXPORT CODE

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```
_id: ObjectId('69d2a63b363c60005a7c2907')
name : "Soma Chatterjee"
rollNo : 101
department : "CSE"
age : 21
```

```
_id: ObjectId('69d2a64d363c60005a7c2908')
name : "Amit Sharma"
rollNo : 102
department : "ECE"
age : 21
```

```
_id: ObjectId('69d2a64d363c60005a7c2909')
name : "Priya Das"
rollNo : 103
department : "CSE"
age : 19
```

```
_id: ObjectId('69d2a64d363c60005a7c290b')
name : "Neha Roy"
rollNo : 105
department : "CSE"
age : 20
```